

Snapmaker / 3-in-1 3D Printer / Laser Engraving and Cutting

# The Definitive Guide to Laser Engraving and Cutting with the 200mW and 1600mW Laser Module

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Your Snapmaker lets you turn your ideas into reality. You can engrave on dozens of materials like paper, plywood, MDF, hardwood, leather, fabric, sticker, and even food like cookie! Keep reading to learn about the recommended Snapmaker settings for different materials.

Snapmaker offers two types of laser modules. The one comes with the Snapmaker Original 3-in-1 3D printer is a 200mW laser module and you can upgrade it with the new 1600mW laser module.

If you wish to achieve optimal engraving or cutting results with other untested materials, you can utilize the **Material Test** feature to conduct your own tests.

- Cork
- Bamboo
- EVA
- Rubber
- Acrylic
- Corrugated Cardboard
- Plywood
- MDF
- Leather
- Sticker
- Colored Card

\* Make sure that you've set the best work origin for laser engraving and cutting. For 1600mW Laser Module, you can use the Fine Tune Work Origin feature to achieve the best cutting result.

\* The recommended settings might not apply to all situations. Take them as a reference, and you can fine-tune the parameters for your materials :)

## **Cork**

## **Bamboo**

| B&W                                                                             | Grayscale                                                                        | Vector                                                                             |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |

| Bamboo    | For 1600mW Laser Module |       |                          |                           |                                                             |
|-----------|-------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------|
| Engrave ✓ |                         |       | Cut ✗                    |                           |                                                             |
|           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                              |
| Engrave   | B&W                     | 90%   | 3000                     | 1000                      | /                                                           |
|           | Grayscale               | 100%  | 3000                     | 1000                      | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |
|           | Vector                  | 90%   | 3000                     | 800                       | /                                                           |

| Bamboo    | For 200mW Laser Module |       |                          |                           |                                                             |
|-----------|------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------|
| Engrave ✓ |                        |       | Cut ✗                    |                           |                                                             |
|           | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                              |
| Engrave   | B&W                    | 90%   | 2000                     | 800                       | /                                                           |
|           | Grayscale              | 90%   | 2000                     | 500                       | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |
|           | Vector                 | 90%   | 2000                     | 500                       | /                                                           |

EVA



| B&W                                                                             | Grayscale                                                                        | Vector                                                                             |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |

| EVA     |           | For 1600mW Laser Module |       |                          |                           |                                                             |            |
|---------|-----------|-------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------|------------|
|         |           | Engrave ✓               |       |                          | Cut ✓                     |                                                             |            |
|         |           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                              |            |
| Engrave |           | B&W                     | 90%   | 3000                     | 2000                      | /                                                           |            |
|         |           | Grayscale               | 60%   | 3000                     | 3000                      | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |            |
|         |           | Vector                  | 90%   | 3000                     | 2000                      | /                                                           |            |
|         | Thickness | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                  | Pass Depth |
| Cut     | 5mm       | Vector                  | 100%  | 2000                     | 90                        | 1                                                           | 0          |

| EVA     |  | For 200mW Laser Module |       |                          |                           |                                                             |  |
|---------|--|------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------|--|
|         |  | Engrave ✓              |       |                          | Cut ✗                     |                                                             |  |
|         |  | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                              |  |
| Engrave |  | B&W                    | 90%   | 2000                     | 1000                      | /                                                           |  |
|         |  | Grayscale              | 90%   | 2000                     | 1000                      | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |  |
|         |  | Vector                 | 90%   | 2000                     | 1000                      | /                                                           |  |

Rubber



| B&W                                                                             | Grayscale                                                                        | Vector                                                                             |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |

| Rubber    | For 1600mW Laser Module |       |                        |                        |                                                             |
|-----------|-------------------------|-------|------------------------|------------------------|-------------------------------------------------------------|
| Engrave ✓ |                         |       | Cut ✗                  |                        |                                                             |
|           | Mode                    | Power | Jog Speed (mm/minute ) | Work Speed (mm/minute) | Other Settings                                              |
| Engrave   | B&W                     | 90%   | 2000                   | 1500                   | /                                                           |
|           | Grayscale               | 90%   | 3000                   | 800                    | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |
|           | Vector                  | 100%  | 2000                   | 500                    | /                                                           |

| Rubber    | For 200mW Laser Module |       |                        |                        |                                                             |
|-----------|------------------------|-------|------------------------|------------------------|-------------------------------------------------------------|
| Engrave ✓ |                        |       | Cut ✗                  |                        |                                                             |
|           | Mode                   | Power | Jog Speed (mm/minute ) | Work Speed (mm/minute) | Other Settings                                              |
| Engrave   | B&W                    | 100%  | 2000                   | 1000                   | /                                                           |
|           | Grayscale              | 100%  | 2000                   | 400                    | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |
|           | Vector                 | 100%  | 2000                   | 400                    | /                                                           |

Acrylic



**Corrugated Cardboard**

| B&W                                                                              | Grayscale                                                                        | Vector                                                                             |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |

| Corrugated Cardboard |           | For 1600mW Laser Module |       |                       |                        |                                                             |            |
|----------------------|-----------|-------------------------|-------|-----------------------|------------------------|-------------------------------------------------------------|------------|
|                      |           | Engrave ✓               |       |                       | Cut ✓                  |                                                             |            |
|                      |           | Mode                    | Power | Jog Speed (mm/minute) | Work Speed (mm/minute) | Other Settings                                              |            |
| Engrave              |           | B&W                     | 80%   | 3000                  | 2000                   | /                                                           |            |
|                      |           | Grayscale               | 50%   | 3000                  | 1000                   | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |            |
|                      |           | Vector                  | 90%   | 3000                  | 800                    | /                                                           |            |
|                      | Thickness | Mode                    | Power | Jog Speed (mm/minute) | Work Speed (mm/minute) | Multi-pass                                                  | Pass Depth |
| Cut                  | 1mm       | Vector                  | 100%  | 1500                  | 180                    | 2                                                           | 0          |
|                      | 3mm       | Vector                  | 100%  | 1500                  | 130                    | 2                                                           | 0          |
|                      | 4mm       | Vector                  | 100%  | 1500                  | 120                    | 3                                                           | 1          |

| Corrugated Cardboard |  | For 200mW Laser Module |       |                       |                        |                                                             |  |
|----------------------|--|------------------------|-------|-----------------------|------------------------|-------------------------------------------------------------|--|
|                      |  | Engrave ✓              |       |                       | Cut ✗                  |                                                             |  |
|                      |  | Mode                   | Power | Jog Speed (mm/minute) | Work Speed (mm/minute) | Other Settings                                              |  |
| Engrave              |  | B&W                    | 90%   | 2000                  | 1000                   | /                                                           |  |
|                      |  | Grayscale              | 100%  | 2000                  | 800                    | Algorithm: Atkinson<br>Movement Mode: Line (Normal Quality) |  |
|                      |  | Vector                 | 90%   | 2000                  | 500                    | /                                                           |  |



Plywood

| Plywood                              | For 200mW Laser Module |       |                          |                           |                                                                                     |
|--------------------------------------|------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------------------------------|
| Engrave ✓                      Cut ✗ |                        |       |                          |                           |                                                                                     |
|                                      | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                      |
| Engrave                              | B&W                    | 100%  | 1000                     | 1000                      | /                                                                                   |
|                                      | Grayscale              | 100%  | 1500                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 30 |
|                                      | Vector                 | 100%  | 1000                     | 1000                      | /                                                                                   |

| Plywood |           | For 1600mW Laser Module |       |                          |                           |                                                                                    |            |
|---------|-----------|-------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|------------|
|         |           | Engrave ✓               |       |                          | Cut ✓                     |                                                                                    |            |
|         |           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |            |
| Engrave |           | B&W                     | 100%  | 1500                     | 1500                      | /                                                                                  |            |
|         |           | Grayscale               | 100%  | 1500                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 5 |            |
|         |           | Vector                  | 100%  | 1500                     | 1500                      | /                                                                                  |            |
|         | Thickness | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                         | Pass Depth |
| Cut     | 1.5mm     | Vector                  | 100%  | 1500                     | 75                        | 1                                                                                  | 0          |
|         | 2.2mm     | Vector                  | 100%  | 1500                     | 120                       | 3                                                                                  | 0.4        |
|         | 2.8mm     | Vector                  | 100%  | 1500                     | 120                       | 4                                                                                  | 0.4        |

MDF



| MDF                                  | For 200mW Laser Module |       |                          |                           |                                                                                     |
|--------------------------------------|------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------------------------------|
| Engrave ✓                      Cut ✗ |                        |       |                          |                           |                                                                                     |
|                                      | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                      |
| Engrave                              | B&W                    | 100%  | 1000                     | 1000                      | /                                                                                   |
|                                      | Grayscale              | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 10 |
|                                      | Vector                 | 100%  | 1000                     | 1000                      | /                                                                                   |

| MDF            | For 1600mW Laser Module |       |                          |                           |                                                                                    |
|----------------|-------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|
| Engrave ✓Cut ✗ |                         |       |                          |                           |                                                                                    |
|                | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |
| Engrave        | B&W                     | 100%  | 2000                     | 2000                      | /                                                                                  |
|                | Grayscale               | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 2 |
|                | Vector                  | 100%  | 2000                     | 2000                      | /                                                                                  |

Leather



| Leather | For 200mW Laser Module |           |                          |                           |                                                                                     |
|---------|------------------------|-----------|--------------------------|---------------------------|-------------------------------------------------------------------------------------|
|         |                        | Engrave ✓ |                          | Cut ✕                     |                                                                                     |
|         | Mode                   | Power     | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                      |
| Engrave | B&W                    | 100%      | 1500                     | 400                       | /                                                                                   |
|         | Grayscale              | 100%      | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 15 |
|         | Vector                 | 100%      | 1500                     | 400                       | /                                                                                   |

| Leather |           | For 1600mW Laser Module |       |                          |                           |                                                                                    |            |
|---------|-----------|-------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|------------|
|         |           | Engrave ✓               |       |                          | Cut ✓                     |                                                                                    |            |
|         |           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |            |
| Engrave |           | B&W                     | 70%   | 2000                     | 2000                      | /                                                                                  |            |
|         |           | Grayscale               | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 3 |            |
|         |           | Vector                  | 70%   | 1500                     | 300                       | /                                                                                  |            |
|         | Thickness | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                         | Pass Depth |
| Cut     | 0.6mm     | Vector                  | 70%   | 1500                     | 300                       | 2                                                                                  | 0          |

**Sticker**



| Sticker |           | For 200mW Laser Module |       |                          |                           |                                                                                    |            |
|---------|-----------|------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|------------|
|         |           | Engrave ✓              |       |                          | Cut ✓                     |                                                                                    |            |
|         |           | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |            |
| Engrave |           | B&W                    | 100%  | 2000                     | 1400                      | /                                                                                  |            |
|         |           | Grayscale              | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 3 |            |
|         |           | Vector                 | 60%   | 2000                     | 2000                      | /                                                                                  |            |
|         | Thickness | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                         | Pass Depth |
| Cut     | 0.3mm     | Vector                 | 100%  | 1500                     | 300                       | 1                                                                                  | 0          |

| Sticker |           | For 1600mW Laser Module |       |                          |                           |                                                                                    |            |
|---------|-----------|-------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|------------|
|         |           | Engrave ✓               |       |                          | Cut ✓                     |                                                                                    |            |
|         |           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |            |
| Engrave |           | B&W                     | 40%   | 2000                     | 2000                      | /                                                                                  |            |
|         |           | Grayscale               | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 3 |            |
|         |           | Vector                  | 35%   | 1500                     | 300                       | /                                                                                  |            |
|         | Thickness | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                         | Pass Depth |
| Cut     | 0.3mm     | Vector                  | 100%  | 1500                     | 800                       | 1                                                                                  | 0          |

Colored Card



| Colored Card |           | For 200mW Laser Module |       |                          |                           |                                                                                     |            |
|--------------|-----------|------------------------|-------|--------------------------|---------------------------|-------------------------------------------------------------------------------------|------------|
|              |           | Engrave ✓              |       |                          | Cut ✓                     |                                                                                     |            |
|              |           | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                      |            |
| Engrave      |           | B&W                    | 100%  | 1500                     | 300                       | /                                                                                   |            |
|              |           | Grayscale              | 100%  | 2000                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 15 |            |
|              |           | Vector                 | 100%  | 1500                     | 300                       | /                                                                                   |            |
|              | Thickness | Mode                   | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                          | Pass Depth |
| Cut          | 0.3mm     | Vector                 | 100%  | 1500                     | 200                       | 1                                                                                   | 0          |

| Colored Card |           | For 1600mW Laser Module |       |                          |                           |                                                                                    |            |
|--------------|-----------|-------------------------|-------|--------------------------|---------------------------|------------------------------------------------------------------------------------|------------|
|              |           | Engrave ✓               |       |                          | Cut ✓                     |                                                                                    |            |
|              |           | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Other Settings                                                                     |            |
| Engrave      |           | B&W                     | 70%   | 2000                     | 2000                      | /                                                                                  |            |
|              |           | Grayscale               | 70%   | 1500                     | /                         | Algorithm: Atkinson<br>Movement Mode: Dot (High Quality)<br>Dwell time (ms/dot): 4 |            |
|              |           | Vector                  | 40%   | 2000                     | 2000                      | /                                                                                  |            |
|              | Thickness | Mode                    | Power | Jog Speed<br>(mm/minute) | Work Speed<br>(mm/minute) | Multi-pass                                                                         | Pass Depth |
| Cut          | 0.3mm     | Vector                  | 40%   | 1500                     | 600                       | 1                                                                                  | 0          |

Feel free to contact us for more help at [support@snapmaker.com](mailto:support@snapmaker.com). We are always available to help you out.

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